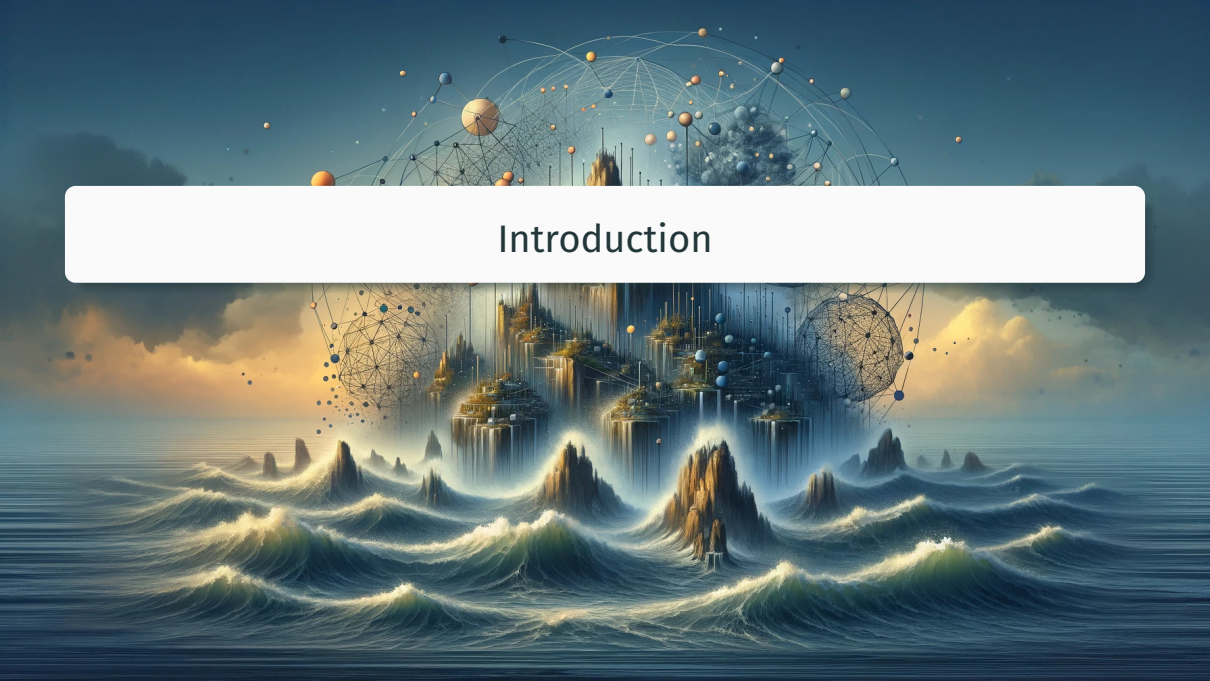


The Complexity of Homomorphism Reconstructibility

Nina Runde

STACS 2024

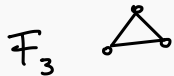
Joint work with Jan Böker, Louis Härtel, Tim Seppelt and Christoph Standke



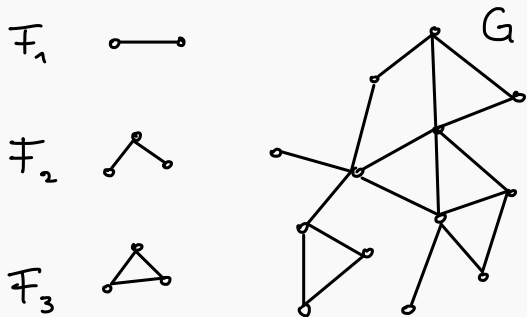
Introduction

Homomorphism Embedding

Homomorphism Embedding



Homomorphism Embedding

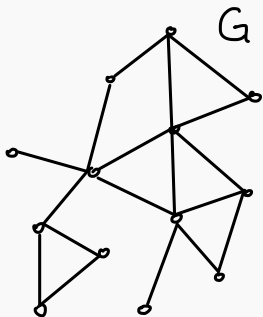


Homomorphism Embedding

F_1 

F_2 

F_3 



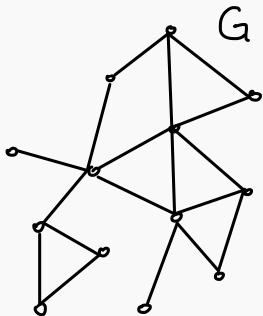
homomorphism
from F to G

Homomorphism Embedding

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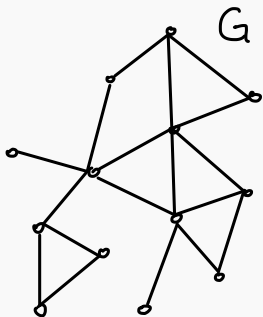
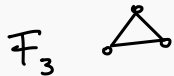
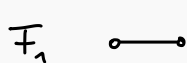
F_2 

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homomorphism
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 $h: V(F) \rightarrow V(G)$

Homomorphism Embedding



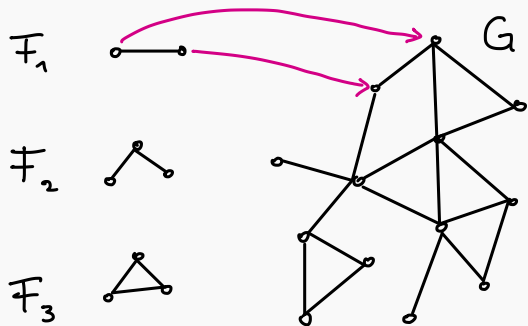
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$$(u, v) \in E(F) \text{ then } (h(u), h(v)) \in E(G)$$

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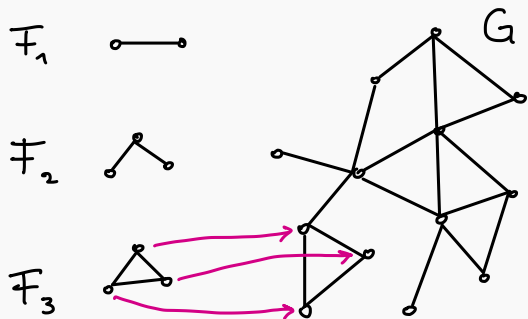
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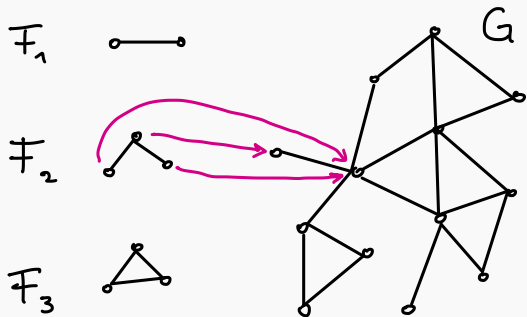
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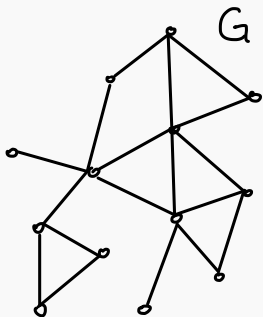
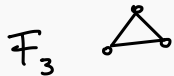
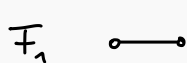
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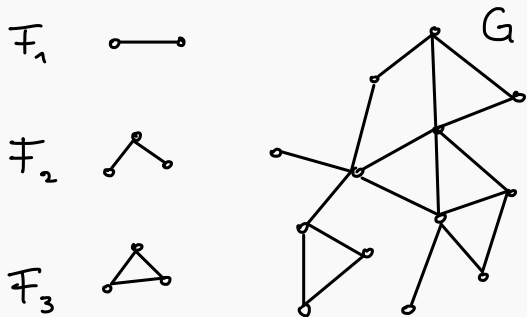
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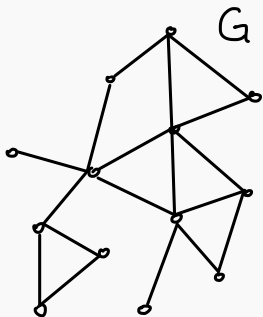
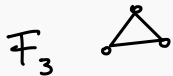
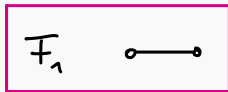
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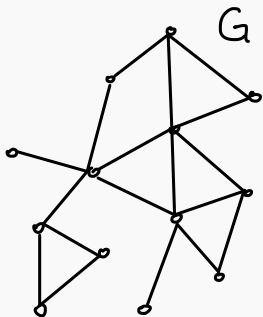
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Homomorphism Embedding

F_1 

F_2 

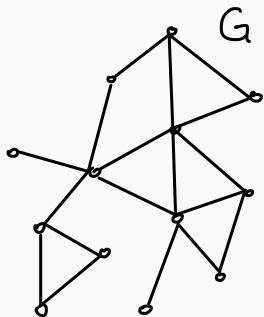
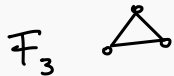
F_3 



$$\text{hom}(F_1, G) = 36$$

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Homomorphism Embedding



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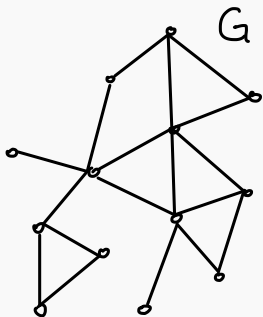
$$\text{hom}(F_3, G) = 30$$

Homomorphism Embedding

$$F_1 \quad \text{---}$$

$$F_2 \quad \begin{array}{c} \circ \\ \diagup \quad \diagdown \\ \circ \quad \circ \end{array}$$

$$F_3 \quad \begin{array}{c} \circ \\ \diagup \quad \diagdown \\ \circ \quad \circ \\ \diagdown \quad \diagup \\ \circ \end{array}$$



$$\text{hom}(F_1, G) = 36$$

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$$G \longrightarrow \mathbb{N}^3$$

Homomorphism Reconstruction

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Input:

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Input:

$$\mathbb{F}_1 \quad h_1 \in \mathbb{N}$$

$$\mathbb{F}_2 \quad h_2 \in \mathbb{N}$$

$$\mathbb{F}_3 \quad h_3 \in \mathbb{N}$$

(...)

Homomorphism Reconstruction

Input:

$\mathbb{F}_1$

$h_1 \in \mathbb{N}$

$\mathbb{F}_2$

$h_2 \in \mathbb{N}$

$\mathbb{F}_3$

$h_3 \in \mathbb{N}$

$(\dots)$

Question:

Homomorphism Reconstruction

Input:

F_1

$h_1 \in \mathbb{N}$

F_2

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(...)

Question:

Is there G with:

$$- \text{hom}(F_1, G) = h_1$$

$$- \text{hom}(F_2, G) = h_2$$

$$- \text{hom}(F_3, G) = h_3$$

(...)

Homomorphism Reconstruction

Input:

F_1  $h_1 \in \mathbb{N}$

F_2  $h_2 \in \mathbb{N}$

F_3  $h_3 \in \mathbb{N}$

Question:

Is there G with:

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Homomorphism Reconstruction

Input:

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$$\mathbb{F}_1 \quad \text{---} \quad h_1 = 36$$

$$\mathbb{F}_2 \quad \text{---} \quad h_2 = 124$$

$$\mathbb{F}_3 \quad \text{---} \quad h_3 = 30$$

Homomorphism Reconstruction

Input:

$$\mathbb{F}_1 \quad \text{---} \quad h_1 = 36$$

Yes!

$$\mathbb{F}_2 \quad \text{---} \quad h_2 = 124$$

$$\mathbb{F}_3 \quad \text{---} \quad h_3 = 30$$

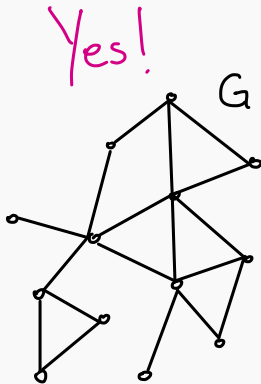
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$$F_3 \quad \text{---} \quad h_3 = 30$$



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Homomorphism Reconstruction

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$$F_1 \quad h_1 \in \mathbb{N}$$

$$F_2 \quad h_2 \in \mathbb{N}$$

$$F_3 \quad h_3 \in \mathbb{N}$$

(...)

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Is there G with:

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(...)

Homomorphism Reconstruction

Input:

$$F_1 \quad h_1 \in \mathbb{N}$$

$$F_2 \quad h_2 \in \mathbb{N}$$

$$F_3 \quad h_3 \in \mathbb{N}$$

$$(\dots) \quad F_i \in \mathcal{F}$$

Question:

Is there G with:

$$- \text{hom}(F_1, G) = h_1$$

$$- \text{hom}(F_2, G) = h_2$$

$$- \text{hom}(F_3, G) = h_3$$

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Homomorphism Reconstruction

Input:

F_1

$h_1 \in \mathbb{N}$

F_2

$h_2 \in \mathbb{N}$

F_3

$h_3 \in \mathbb{N}$

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Is there G with:

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- $\text{hom}(F_2, G) = h_2$

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$(\dots) G \in \mathcal{G}$

Problem: $\text{HOMREC}(\mathcal{F}, \mathcal{G})$

Homomorphism Reconstruction

Input:

F_1

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F_2

$h_2 \in \mathbb{N}$

F_3

$h_3 \in \mathbb{N}$

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- **Constructive:** Explicitly construct a graph with the required count
 - not polynomial in the (binary-)encoding of the homomorphism numbers.
 - Additional problem $\text{BHOMREC}(\mathcal{F}, \mathcal{G})$, where a bound on the number of vertices in G is encoded as unary.

Corollary

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Lemma

If the constraints (F_i, h_i) are realised by some G then there exists H realising them such that

$$|V(H)| \leq \sum h_i |V(F_i)|.$$

Decidability

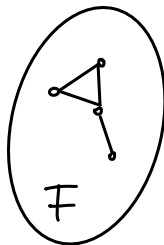
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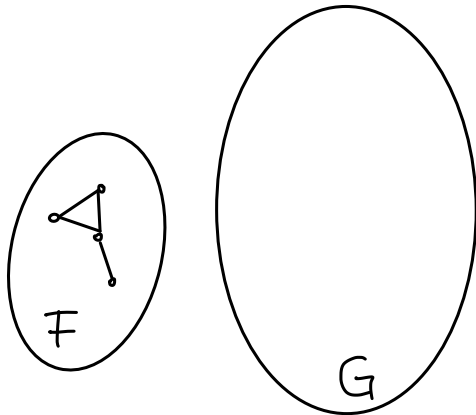
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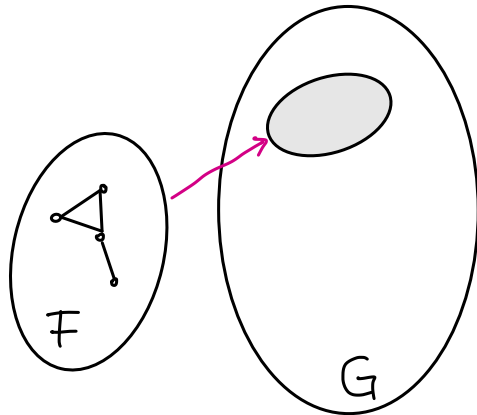
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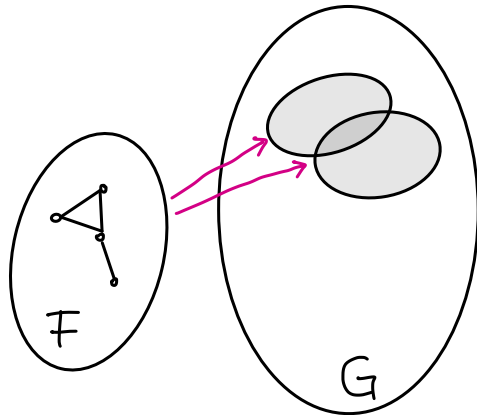
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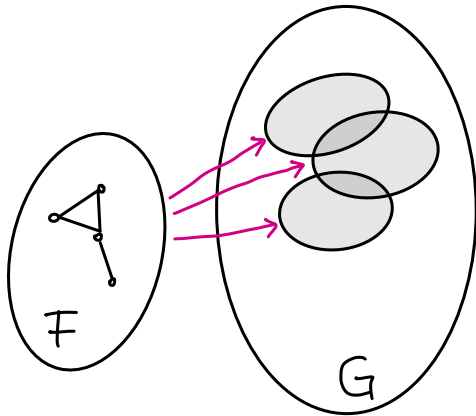
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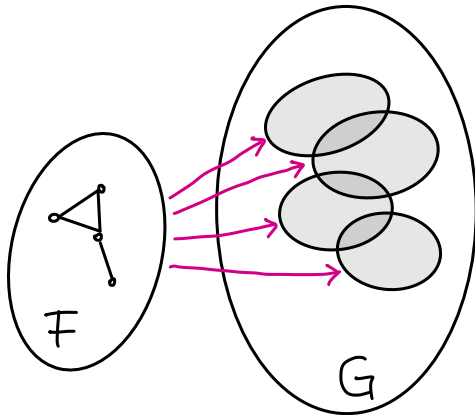
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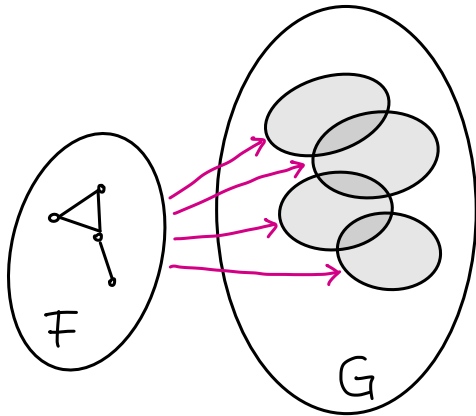
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Corollary

HOMREC is decidable.





The Ocean of Hardness

Hardness Results

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Let $\mathcal{F}$ denote the class of *all graphs*. Then, $\text{HOMREC}(\mathcal{F})$ and $\text{BHOMREC}(\mathcal{F})$ are $\text{NP}^{\text{C=P}}$ -hard.

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Theorem

There is a *finite set* $\mathcal{F}$ of graphs such that $\text{HOMREC}(\mathcal{F})$ is NP-hard. If $\text{BHOMREC}(\mathcal{F})$ is NP-complete for a finite set $\mathcal{F}$ of graphs, then $\text{P} = \text{NP}$.

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The problem

EC-3-COLOURING:

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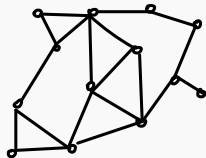
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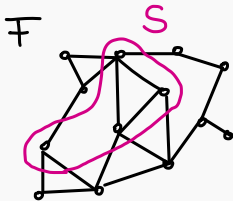
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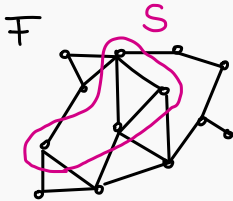
Theorem

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The problem

EC-3-COLOURING:

Input : $(\mathbb{F}, \textcolor{violet}{S}, \textcolor{violet}{k})$



$\textcolor{violet}{k} \in \mathbb{N}$

HOMREC($\mathcal{F}$) is $\text{NP}^{\text{C=P}}$ -hard

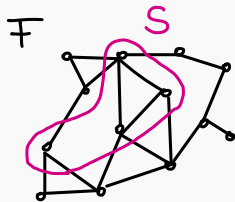
Theorem

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The problem

EC-3-COLOURING:

Input : $(\mathbb{F}, \textcolor{violet}{S}, \textcolor{teal}{k})$ Question:



HOMREC($\mathcal{F}$) is $\text{NP}^{\text{C=P}}$ -hard

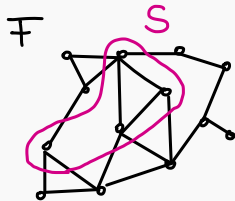
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Let $\mathcal{F}$ be the class of all labelled graphs. Then, $\text{EC-3-COLOURING} \leq_p \text{HOMREC}(\mathcal{F})$.

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Input : $(\mathcal{F}, S, k)$



$k \in \mathbb{N}$

Question:

Is there a
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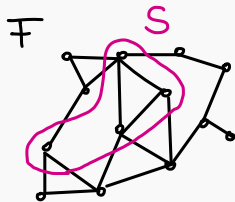
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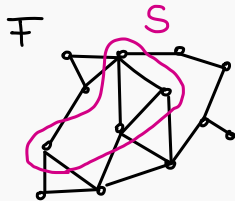
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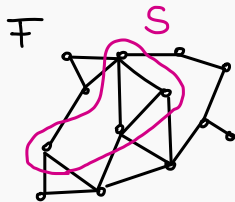
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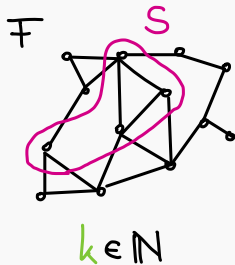
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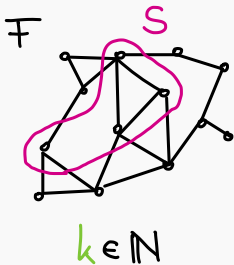
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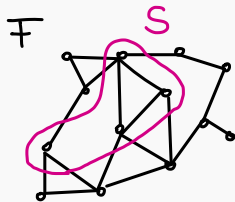
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Input:

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$(\text{---}o, 6)$

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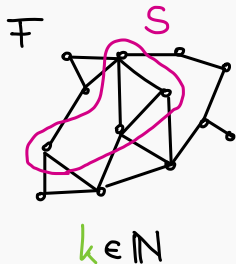
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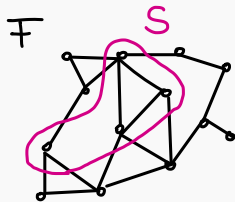
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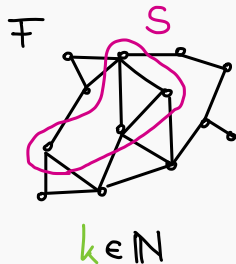
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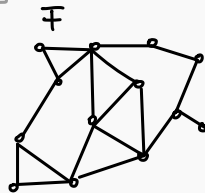
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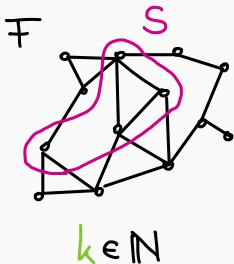
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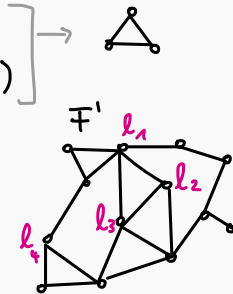
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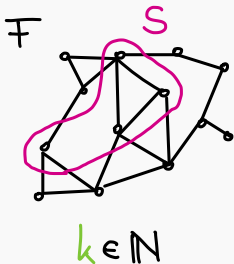
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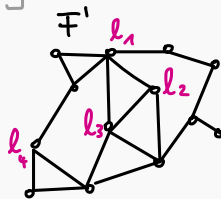
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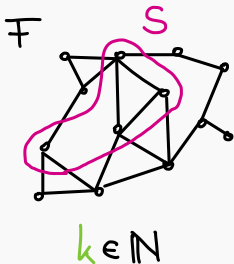
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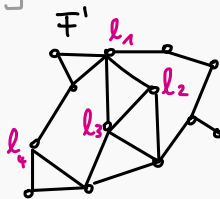
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$(\mathcal{F}', k)$

$(\circ_{l_i}, 1)$



HOMREC($\mathcal{F}$) is NP-hard for finite $\mathcal{F}$

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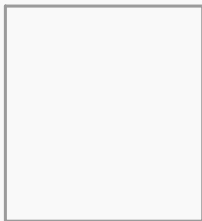
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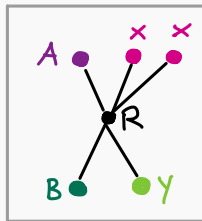
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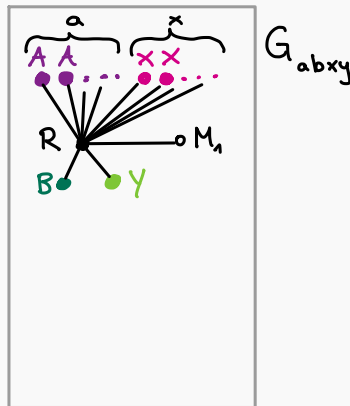
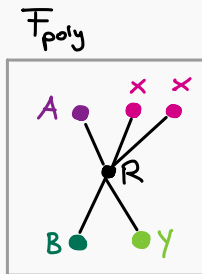
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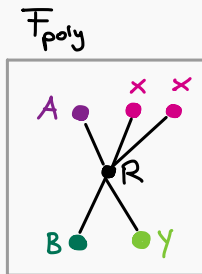
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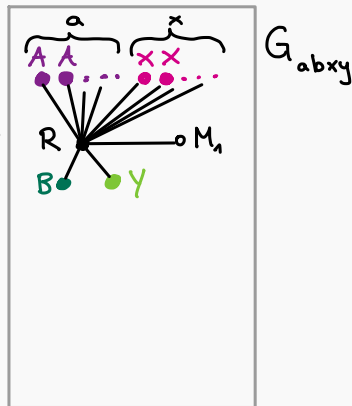
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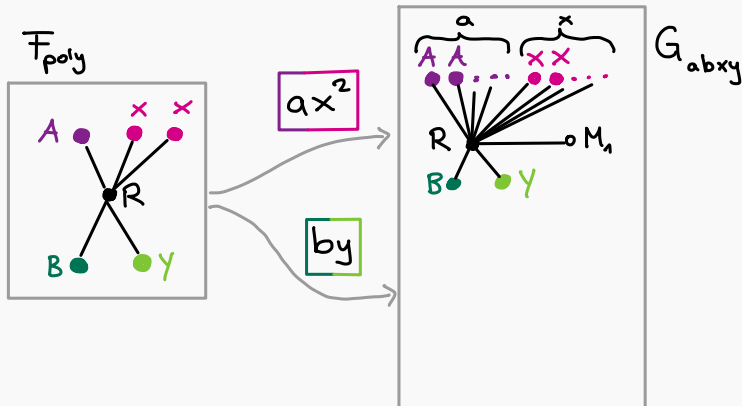
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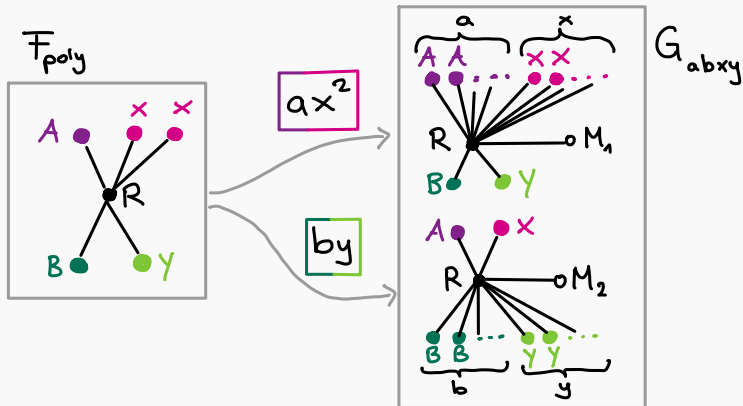
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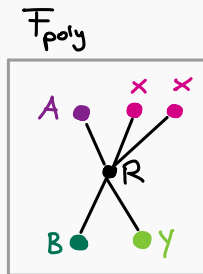
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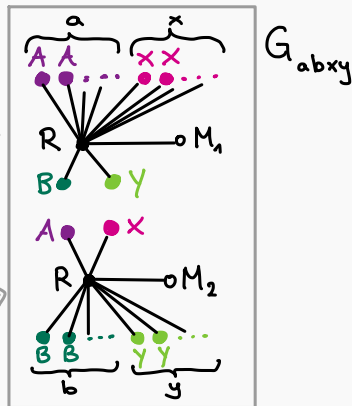
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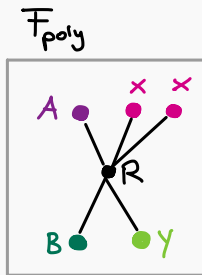
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$$\begin{aligned} \text{hom}(\overset{A}{\bullet} \text{---} \overset{R}{\bullet}) &= a+1 \\ \text{hom}(\overset{B}{\bullet} \text{---} \overset{R}{\bullet}) &= b+1 \\ \text{hom}(\bullet \text{---} \overset{R}{\bullet}) &= 2 \\ \text{hom}(\overset{M_1}{\circ} \text{---} \overset{R}{\bullet}) &= 1 \\ \text{hom}(\overset{M_2}{\circ} \text{---} \overset{R}{\bullet}) &= 1 \\ \text{hom}(\overset{M_2}{\circ} \text{---} \overset{R}{\bullet} \text{---} \overset{M_1}{\circ}) &= 0 \\ \text{hom}(\overset{M_1}{\circ} \text{---} \overset{R}{\bullet} \text{---} \overset{y}{\bullet}) &= 1 \\ \text{hom}(\overset{M_2}{\circ} \text{---} \overset{R}{\bullet} \text{---} \overset{x}{\bullet}) &= 1 \end{aligned}$$

G_{abxy}

Hardness Results

Theorem

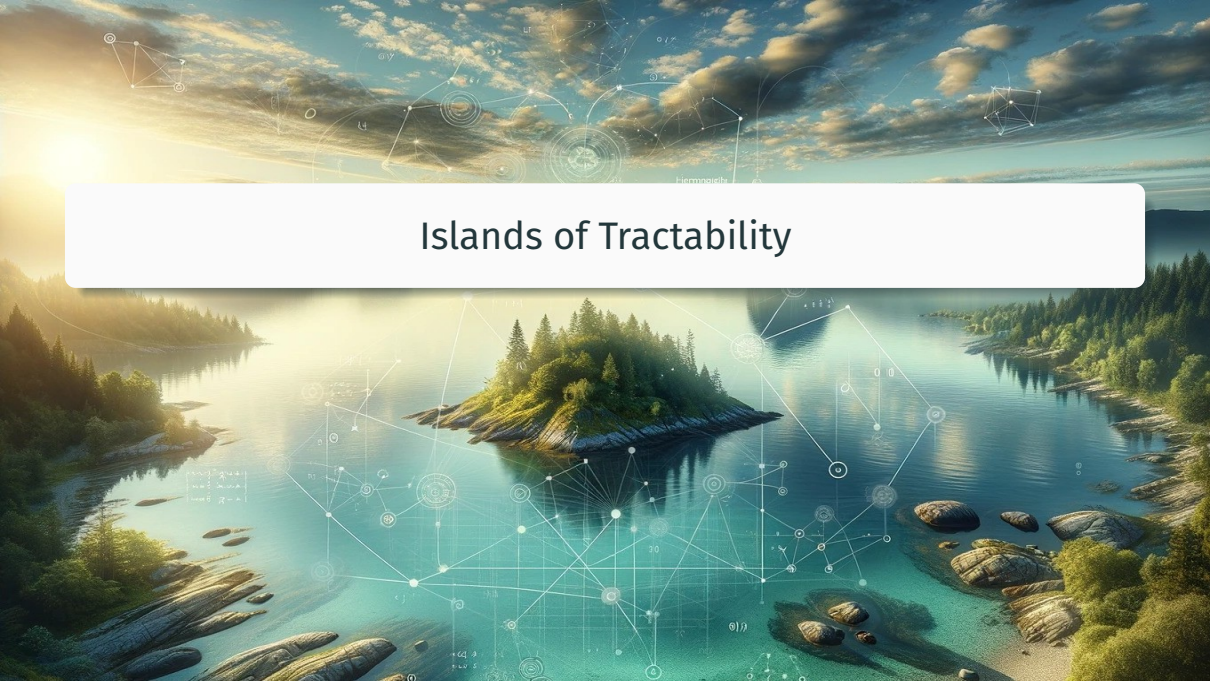
Let $\mathcal{F}$ denote the class of *all graphs*. Then, $\text{HOMREC}(\mathcal{F})$ and $\text{BHOMREC}(\mathcal{F})$ are $\text{NP}^{\text{C=P}}$ -hard.

Theorem

There is a class $\mathcal{F}$ of graphs of *bounded treewidth* such that $\text{HOMREC}(\mathcal{F})$ and $\text{BHOMREC}(\mathcal{F})$ are NP-hard.

Theorem

There is a *finite set* $\mathcal{F}$ of graphs such that $\text{HOMREC}(\mathcal{F})$ is NP-hard. If $\text{BHOMREC}(\mathcal{F})$ is NP-complete for a finite set $\mathcal{F}$ of graphs, then $\text{P} = \text{NP}$.



Islands of Tractability

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 - FPT for HOMREC with a single connected constraint graph,
 - FPT for SUBREC where all constraint graphs are connected and have the same size.

From Small to Big

Building blocks:

$$S(\mathcal{I}) := \{\text{hom}(\mathcal{I}, G) \in \mathbb{N}^{\mathcal{I}} \mid G \text{ graph s.t. } |G| \leq k\}.$$

From Small to Big

Building blocks:

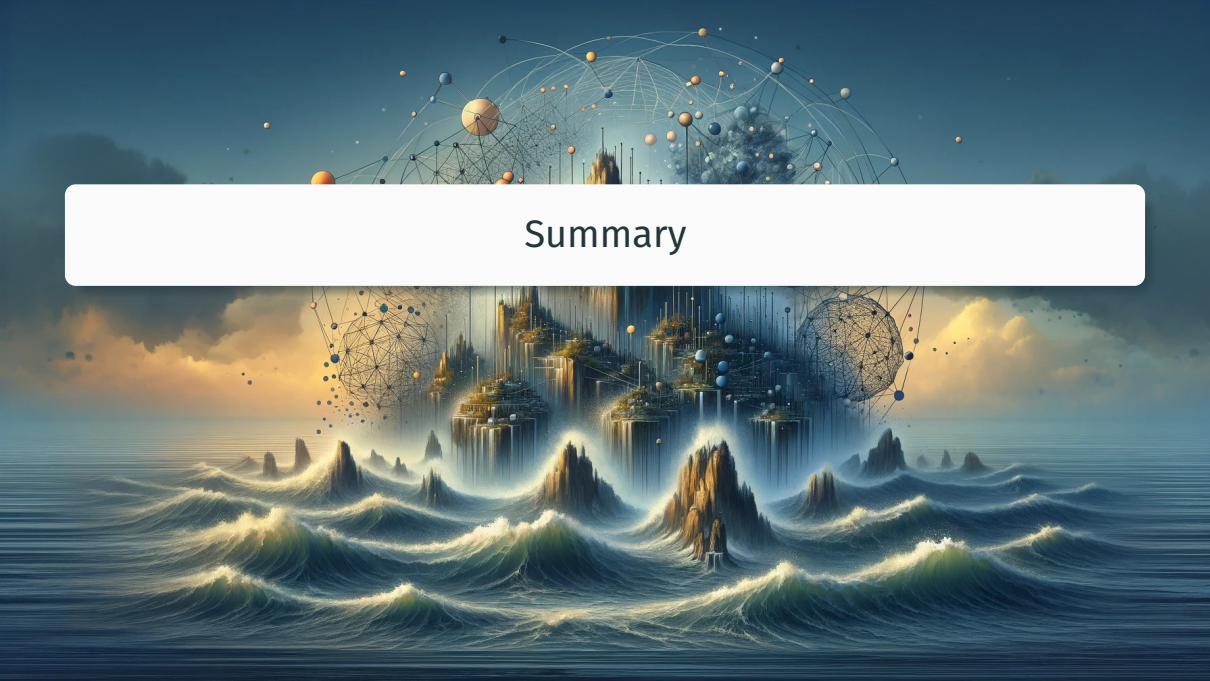
$$S(\mathcal{I}) := \{\text{hom}(\mathcal{I}, G) \in \mathbb{N}^{\mathcal{I}} \mid G \text{ graph s.t. } |G| \leq k\}.$$

Lemma

If F is connected then

$$\text{hom}(F, G_1 + G_2) = \text{hom}(F, G_1) + \text{hom}(F, G_2)$$

$$\text{sub}(F, G_1 + G_2) = \text{sub}(F, G_1) + \text{sub}(F, G_2)$$



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all	$\text{NP}^{\text{C}=\text{P}}$ -hard	
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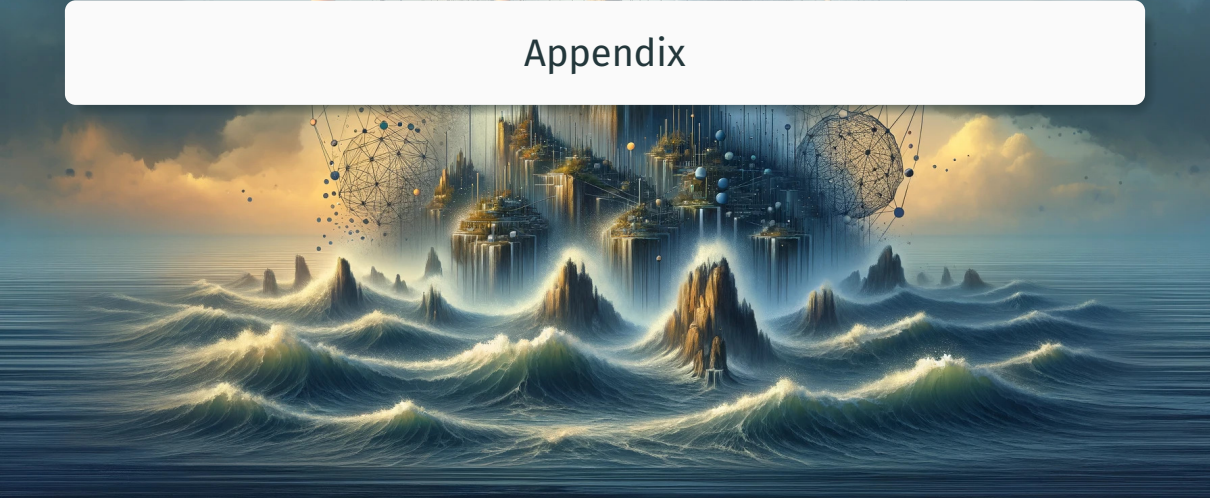
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- Result for subgraph count of cliques
- Is $p\text{-HOMREC}$ with two connected constraints tractable?



Appendix



The background image is a surreal digital artwork. The top half shows a dark blue sky with a complex network of nodes and lines forming a globe-like structure. A cityscape is visible in the background, and a large sphere of light is on the right. The bottom half shows a turbulent sea with large, white-capped waves. In the distance, a cityscape is built on a series of floating, tiered platforms. The platforms are connected by a network of lines and nodes, similar to the one in the top half. A large sphere of light is also visible on the right side of the platforms.

Problem Definition

Problem (HOMREC)

INPUT: Pairs (F_i, h_i) where F_i is a graph and $h_i \in \mathbb{N}$

QUESTION: Does there exist G such that $\text{hom}(F_i, G) = h_i$ for all i ?

Problem (BHOMREC)

INPUT: An integer $N \in \mathbb{N}$, pairs (F_i, h_i) where F_i is a graph and $h_i \in \mathbb{N}$

QUESTION: Does there exist G with $|G| \leq N$ such that $\text{hom}(F_i, G) = h_i$ for all i ?

Observation

$BHOMREC \leq HOMREC$.

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- Two sources of hardness:
 - Reconstruction hardness
 - Counting hardness